

# Clébio Júnior

## AI ENGINEER

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## PROFESSIONAL SUMMARY

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AI Engineer and Data Scientist with over 5 years of experience developing artificial intelligence, machine learning and NLP solutions applied to real business problems. Hands-on experience building LLM applications, including RAG (Retrieval-Augmented Generation) systems with hybrid search, embeddings, FAISS vector stores and workflow orchestration with LangChain and LangGraph. Practical background in ingestion and processing pipelines for unstructured documents using OCR and NLP, as well as end-to-end predictive modeling from the business problem to API deployment. Master's degree in Natural Sciences, with a solid quantitative background and a focus on measurable, reliable solutions.

## TECHNICAL SKILLS

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- **Generative AI and LLMs:** RAG (Retrieval-Augmented Generation), LangChain, LangGraph, Ollama, Local and API-based LLMs, Prompt Engineering, Embeddings, Sentence Transformers, Semantic Search, Hybrid Search (BM25 + vector), Cross-Encoder Reranking, Document Chunking and Indexing, Agents and Stateful Workflows
- **Vector Stores and Retrieval:** FAISS, Vector Indexing, BM25, Retrieval System Evaluation
- **NLP:** spaCy, NLTK, Text Classification, Named Entity Recognition (NER), Tokenization, Lemmatization, Text Summarization, OCR (Tesseract / pytesseract), pdfplumber, Regular Expressions (Regex)
- **Machine Learning:** scikit-learn, XGBoost, LightGBM, CatBoost, Supervised and Unsupervised Learning, Predictive Modeling, Feature Engineering, Hyperparameter Optimization (Optuna), Interpretability (SHAP)
- **Programming Languages:** Python, SQL, C/C++
- **Engineering and Deployment:** Flask, REST APIs, Docker, Git, GitHub, Linux, MLOps, Model Versioning and Serving
- **Data and Analysis:** pandas, NumPy, SciPy, Exploratory Data Analysis, Time Series, Applied Statistics, SAS, Jupyter Notebook

## PROFESSIONAL EXPERIENCE

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### Data Scientist | AI and NLP Solutions — Vert Analytics

October/2024 – Present | Remote

- Developed an NLP-based recommendation system for legal defenses using embeddings and FAISS vector search, automatically suggesting the most suitable defense for each legal opinion, reducing manual research time and increasing decision accuracy.
- Built ingestion and structuring pipelines for unstructured PDF documents, including scanned files, combining pdfplumber, Tesseract OCR and regular expressions to turn raw text into analyzable data.
- Implemented a text classification and analysis solution with spaCy to categorize customer complaints on social media, applying tokenization, lemmatization, stopword removal and normalization to generate strategic insights for product and customer service.
- Conduct time series analysis and anomaly detection over the extracted data, delivering results through interactive SAS dashboards.

### Data Scientist — Datarisk

January/2022 – August/2024 | Remote

- Developed a machine learning credit scoring model to estimate repayment probability from customers' financial history, improving credit approval criteria and the efficiency of risk management.
- Built a regression model for sales volume forecasting, with preprocessing, feature engineering and hyperparameter optimization, which went on to support strategic planning for the commercial team.
- Applied clustering algorithms for customer base segmentation, identifying distinct behavioral profiles and enabling targeted relationship strategies.
- Worked across the full model lifecycle, from framing the business problem to validating results with business stakeholders.

### Data Scientist — Be.X!

March/2021 – January/2022 | Remote

- Developed machine learning algorithms to automatically generate insights from clients' structured and unstructured data.
- Processed large volumes of unstructured text data with regular expressions for cleaning and information extraction, significantly increasing data availability for decision-making.
- Implemented rule-based outlier detection algorithms to flag operational anomalies, contributing to risk mitigation.

- Developed a predictive model for delivery delay probability, using ranking-based metrics to prioritize the highest-risk cases and enable preventive actions in logistics operations.

## SELECTED PROJECTS

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- **Legal Decision RAG — Hybrid RAG System for Legal Documents** — Retrieval-Augmented Generation system for natural language queries over administrative decisions on fine appeals. Full architecture with PDF ingestion and chunking, embedding generation, FAISS vector index, hybrid retrieval (BM25 + semantic search), cross-encoder reranking and LangGraph orchestration, including query routing, decomposition of complex questions, rewriting of ambiguous queries and a critique loop over the generated answers. Runs on local LLMs via Ollama. ([github.com/juniorcl/legal-decision-rag](https://github.com/juniorcl/legal-decision-rag))
- **DSToolkit — Python Data Science Library** — Modular library with a scikit-learn-inspired API, bringing together AutoML pipelines for classification, regression and clustering, hyperparameter optimization, evaluation metrics (ROC, PR, KS, calibration curves) and interpretability and feature engineering tools. ([github.com/juniorcl/data-science-toolkit](https://github.com/juniorcl/data-science-toolkit))
- **E-commerce Delivery Time Forecast (Olist)** — End-to-end ML pipeline with data consumed via SQL from a Docker-containerized database, feature selection with mRMR, a LightGBM model optimized with Optuna and interpretability with SHAP. Achieved a 66% MAE reduction over the existing business rule, deployed as a Flask API. ([github.com/juniorcl/olist-delivery-forecast](https://github.com/juniorcl/olist-delivery-forecast))
- **Transaction Fraud Detection** — Classification model on highly imbalanced data with feature engineering, feature selection using Boruta and hyperparameter tuning. The final XGBoost model achieved 91.5% balanced accuracy and 94.4% precision on unseen data. Project with over 220 stars on GitHub. ([github.com/juniorcl/transaction-fraud-detection](https://github.com/juniorcl/transaction-fraud-detection))
- **Text Recognition with OCR** — Applied study on extracting text from images and scanned documents with Tesseract and Python, the technical foundation for the document pipelines used in production.

Full portfolio available at [juniorcl.github.io](https://juniorcl.github.io) and [github.com/juniorcl](https://github.com/juniorcl).

## EDUCATION

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### Master's Degree in Natural Sciences — Environmental Science

Universidade Estadual do Norte Fluminense (UENF) | 2017 – 2019

### Bachelor's Degree in Natural Sciences — Physics

Instituto Federal Fluminense (IFF) | 2010 – 2016

## CERTIFICATIONS AND COURSES

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- Quickstart: LangGraph Essentials — Python
- LangChain Academy — LangGraph
- Text Recognition with OCR and Python
- Text Summarization with Natural Language Processing
- Clustering: Extracting Patterns from Data
- MLOps: Model Deployment and Machine Learning APIs
- Data Science Program — Comunidade DS

## LANGUAGES

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- **Portuguese:** Native
- **English:** Full Professional Proficiency
- **Esperanto:** Full Professional Proficiency